

■ 2.1.7 Manipulating Expressions like Lists

Since lists are just a particular kind of expression, it will come as no surprise that you can use most of the list manipulation functions from Section 1.7 on any expression.

Here is an expression that corresponds to a sum of terms.

```
In[1]:= t = 1 + x + x^2 + y^2
```

```
Out[1]= 1 + x + x^2 + y^2
```

`Take[2, t]` takes the first two elements from `t`, just as if `t` were a list.

```
In[2]:= Take[t, 2]
```

```
Out[2]= 1 + x
```

`Length` gives the number of elements in `t`.

```
In[3]:= Length[t]
```

```
Out[3]= 4
```

This extracts the third part of `t`.

```
In[4]:= t[[3]]
```

```
Out[4]= x^2
```

You can use `FreeQ[expr, form]` to test whether `form` appears anywhere in `expr`.

```
In[5]:= FreeQ[t, x]
```

```
Out[5]= False
```

Here is a function with four arguments.

```
In[6]:= f[a, b, c, d]
```

```
Out[6]= f[a, b, c, d]
```

You can add an argument using `Append`.

```
In[7]:= Append[%, e]
```

```
Out[7]= f[a, b, c, d, e]
```

This reverses the arguments.

```
In[8]:= Reverse[%]
```

```
Out[8]= f[e, d, c, b, a]
```

There are a few extra functions that can be used with expressions, as discussed in Sections 2.4.5 and 2.4.6.